Chapter 1. Multivariate Data and the Geometry of the Sample

Data matrices, descriptive statistics, projections, sphering, Mahalanobis distance, eigendecomposition

Contents

1 Multivariate Data and the Goals of Analysis **3**

2 The Data Matrix and Descriptive Statistics **3**

2.1 Data matrix . . . 3

2.2 Sample mean, covariance and correlation . . . 3

2.3 The centering matrix . . . 4

3 Sample Geometry in p-Space **4**

3.1 The data ellipse . . . 4

3.2 Subspaces, orthogonal projection and projection scores . . . 5

3.3 Projection onto an affine subspace and decomposition of observations . . . 5

3.4 Linear and affine transformations . . . 6

3.5 Rotation of coordinates . . . 6

4 Standardisation, Sphering and Mahalanobis Distance **6**

4.1 Univariate standardisation and its limitation . . . 6

4.2 Sphering . . . 7

4.3 Mahalanobis distance . . . 7

5 Eigendecomposition and Positive Definiteness **8**

5.1 Eigendecomposition . . . 8

5.2 Inverse, square root, definiteness . . . 8

5.3 Covariance matrices and ellipses . . . 9

6 Geometry of Principal Component Analysis **9**

6.1 Which direction to project on? . . . 9

6.2 Principal component scores . . . 9

6.3 Dimension reduction and reconstruction error . . . 10

6.4 Proportion of variance, scale and caveats . . . 10

7 Random Vectors and the Geometry of Variable Vectors **11**

7.1 Random vectors . . . 11

7.2 Mean, variance and correlation through variable vectors . . . 11

7.3 Linear combinations of variables . . . 12

8 Dimension and Rank of the Data **12**

1 Multivariate Data and the Goals of Analysis

Multivariate (multidimensional) data are measurements of two or more variables on each individual or experimental unit. The goals of analysis fall into two groups:

(i) **Inference about an unknown population**: understand and infer features of the population from which the data were drawn.

(ii) **Prediction of future observations**: predict the value or category of new observations.

In either case one starts with exploratory data analysis (EDA) to find structure, outliers and relationships among variables. Numerical summaries are descriptive statistics (mean vector, covariance matrix, correlation matrix); graphical summaries are scatterplot matrices, principal component plots, heatmaps, correlation plots and trellis graphs.

Example 1.1: two running examples

Penguins data. For $n = 333$ penguins from the Palmer archipelago there are 4 numerical variables (bill length, bill depth, flipper length, body mass), 3 categorical variables (species, island, sex) and year. Only the numerical variables go into the data matrix $\mathbf{X}$; categorical variables are kept as separate vectors. This is the typical case $n > p$.

Golub gene expression data (1999). For 72 leukemia patients the expression levels of 7129 genes were measured, together with a categorical label ALL/AML. This is high-dimensional, low-sample-size data ($p = 7129 \gg n = 72$). Scientific question: can the leukemia type be predicted from gene expression alone?

2 The Data Matrix and Descriptive Statistics

2.1 Data matrix

Measurements of p variables on n items form an $n \times p$ matrix

$$\mathbf{X} = \begin{pmatrix} x_{11} & \cdots & x_{1p} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{np} \end{pmatrix} = \begin{pmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_n^\top \end{pmatrix} = [\mathbf{x}_{(1)}, \ldots, \mathbf{x}_{(p)}].$$

The ith row $\mathbf{x}_i = (x_{i1}, \ldots, x_{ip})^\top \in \mathbb{R}^p$ is the **observation vector** of item i; the jth column $\mathbf{x}_{(j)} = (x_{1j}, \ldots, x_{nj})^\top \in \mathbb{R}^n$ is the **variable vector** of variable j. All variables in $\mathbf{X}$ are taken to be numerical.

2.2 Sample mean, covariance and correlation

Definition 1.2: descriptive statistics

$$\bar{\mathbf{x}} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{x}_i, \qquad \mathbf{S} = (s_{jk}), \quad s_{jk} = \frac{1}{n-1}\sum_{i=1}^{n}(x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k),$$

$$\mathbf{S} = \frac{1}{n-1}\sum_{i=1}^{n}(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top, \qquad \mathbf{R} = (r_{jk}), \quad r_{jk} = \frac{s_{jk}}{\sqrt{s_{jj}}\sqrt{s_{kk}}}, \qquad \mathbf{R} = \mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2},$$

with $\mathbf{D} = \operatorname{diag}(s_{11}, \ldots, s_{pp})$. $\mathbf{R}$ is defined only when all $s_{jj} > 0$. One also uses $\mathbf{S}_n = \frac{n-1}{n}\mathbf{S}$ (divisor n).

Remark 1.3: a Gaussian with covariance S always exists

$\mathbf{S}$ is symmetric and positive semidefinite, so a symmetric square root $\mathbf{S}^{1/2}$ exists (§5). With $\mathbf{Z} \sim N_p(\mathbf{0}, \mathbf{I}_p)$ and $\mathbf{X} = \boldsymbol{\mu} + \mathbf{S}^{1/2}\mathbf{Z}$, $\mathrm{E}\mathbf{X} = \boldsymbol{\mu}$ and $\mathrm{Cov}\mathbf{X} = \mathbf{S}^{1/2}\mathbf{I}_p\mathbf{S}^{1/2} = \mathbf{S}$.

2.3 The centering matrix

Let $\mathbf{1}_n = (1, \ldots, 1)^\top$, $\mathbf{J}_n = \mathbf{1}_n\mathbf{1}_n^\top$ and $\mathbf{H}_n = \mathbf{I}_n - \frac{1}{n}\mathbf{J}_n$.

Theorem 1.4: matrix forms of $\bar{\mathrm{x}}$ and S

$$\bar{\mathbf{x}} = \tfrac{1}{n}\mathbf{X}^\top\mathbf{1}_n, \qquad \mathbf{X}_C := \mathbf{X} - \mathbf{1}_n\bar{\mathbf{x}}^\top = \mathbf{H}_n\mathbf{X}, \qquad \mathbf{S} = \tfrac{1}{n-1}\mathbf{X}^\top\mathbf{H}_n\mathbf{X} = \tfrac{1}{n-1}\mathbf{X}_C^\top\mathbf{X}_C,$$

and $\mathbf{H}_n^\top = \mathbf{H}_n$, $\mathbf{H}_n^2 = \mathbf{H}_n$, $\mathrm{rank}(\mathbf{H}_n) = n - 1$.

Proof. The jth entry of $\mathbf{X}^\top\mathbf{1}_n$ is $\sum_i x_{ij}$, giving the first identity. Hence $\mathbf{X}_C = \mathbf{X} - \mathbf{1}_n\bar{\mathbf{x}}^\top = \mathbf{X} - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top\mathbf{X} = \mathbf{H}_n\mathbf{X}$. Idempotency follows from $\mathbf{1}_n^\top\mathbf{1}_n = n$:

$$\left(\mathbf{I}_n - \tfrac{1}{n}\mathbf{J}_n\right)^2 = \mathbf{I}_n - \tfrac{2}{n}\mathbf{J}_n + \tfrac{1}{n^2}\mathbf{1}_n(\mathbf{1}_n^\top\mathbf{1}_n)\mathbf{1}_n^\top = \mathbf{I}_n - \tfrac{1}{n}\mathbf{J}_n.$$

Then $\mathbf{X}_C^\top\mathbf{X}_C = \mathbf{X}^\top\mathbf{H}_n^\top\mathbf{H}_n\mathbf{X} = \mathbf{X}^\top\mathbf{H}_n\mathbf{X}$, and $\mathbf{X}_C^\top\mathbf{X}_C = \sum_i(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$. A symmetric idempotent matrix has eigenvalues 0 or 1, so $\mathrm{rank}(\mathbf{H}_n) = \mathrm{tr}(\mathbf{H}_n) = n - 1$. ■

3 Sample Geometry in p-Space

3.1 The data ellipse

Each $\mathbf{x}_i$ is a point in $\mathbb{R}^p$. The mean vector is the centre of gravity of the points and the covariance matrix describes their spread. The **data ellipse**

$$\{\mathbf{x} \in \mathbb{R}^p : (\mathbf{x} - \bar{\mathbf{x}})^\top\mathbf{S}^{-1}(\mathbf{x} - \bar{\mathbf{x}}) = c\}$$

summarises location, shape and size for fixed $c > 0$ (an ellipsoid for $p \geq 3$). The penguin plot uses $c = 5.991 = \chi^2_2(0.05)$. If the cloud is not elliptical, the mean and covariance alone describe it poorly.

Example 1.5: data ellipses

For $\mathbf{S}_1 = \begin{pmatrix} 9 & 0 \\ 0 & 1 \end{pmatrix}$ the axes of the ellipse are the coordinate axes with half-length ratio $3:1$. $\mathbf{S}_2 = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$ has eigenvalues $9, 1$ with eigenvectors $(1,1)^\top/\sqrt{2}$, $(1,-1)^\top/\sqrt{2}$: the same ellipse rotated by 45°. Half-axis lengths are $\sqrt{c\lambda_j}$, so the ellipse reflects the size of variances and covariances.

3.2 Subspaces, orthogonal projection and projection scores

Definition 1.6: subspace and affine subspace

- **Subspace**: $\mathcal{L}_q = \text{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_q\} = \{\mathbf{Vc} : \mathbf{c} \in \mathbb{R}^q\}$, $\mathbf{V} = [\mathbf{v}_1, \ldots, \mathbf{v}_q]$ (linearly independent). It contains the origin.
- **Affine subspace**: $\mathcal{L}_q(\mathbf{x}) = \mathbf{x} + \mathcal{L}_q$, e.g. a line $\{\mathbf{x} + t\mathbf{v}\}$ or a plane $\{\mathbf{x} + c_1\mathbf{v}_1 + c_2\mathbf{v}_2\}$. It is not a vector space unless it contains the origin, but has the same shape as $\mathcal{L}_q$. Its representation (base point, spanning vectors) is not unique.

Definition 1.7: orthogonal projection and projection score

Projection onto $\text{span}\{\mathbf{v}\}$ and the projection score (signed distance):

$$P_{\mathbf{v}}(\mathbf{x}) = \frac{\mathbf{v}\mathbf{v}^\top}{\mathbf{v}^\top\mathbf{v}}\mathbf{x} = s_{\mathbf{v}}(\mathbf{x})\frac{\mathbf{v}}{\sqrt{\mathbf{v}^\top\mathbf{v}}}, \qquad s_{\mathbf{v}}(\mathbf{x}) = \frac{\mathbf{v}^\top\mathbf{x}}{\sqrt{\mathbf{v}^\top\mathbf{v}}}.$$

The projection matrix onto $\mathcal{L}_q$ is $\mathbf{P_V} = \mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top$; for an orthonormal basis ($\mathbf{V}^\top\mathbf{V} = \mathbf{I}_q$)

$$P_{\mathcal{L}_q}(\mathbf{x}) = \mathbf{V}\mathbf{V}^\top\mathbf{x} = s_{\mathbf{v}_1}(\mathbf{x})\mathbf{v}_1 + \cdots + s_{\mathbf{v}_q}(\mathbf{x})\mathbf{v}_q,$$

so $\mathbf{V}^\top\mathbf{x} = (s_{\mathbf{v}_1}(\mathbf{x}), \ldots, s_{\mathbf{v}_q}(\mathbf{x}))^\top$ are the coordinates in the subspace. The projection does not depend on the basis; the scores do.

Derivation: why $\mathbf{P_V}$ is the orthogonal projection

$\mathbf{P_V}^\top = \mathbf{P_V}$ and $\mathbf{P_V}^2 = \mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top\mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top = \mathbf{P_V}$. Also $\mathbf{P_V x} \in \mathcal{L}_q$, and the residual satisfies $\mathbf{V}^\top(\mathbf{x} - \mathbf{P_V x}) = \mathbf{0}$, so it is orthogonal to $\mathcal{L}_q$. By Pythagoras, for any $\mathbf{Vc}$, $\|\mathbf{x} - \mathbf{Vc}\|^2 = \|\mathbf{x} - \mathbf{P_V x}\|^2 + \|\mathbf{P_V x} - \mathbf{Vc}\|^2 \geq \|\mathbf{x} - \mathbf{P_V x}\|^2$: $\mathbf{P_V x}$ is the closest point of $\mathcal{L}_q$ to $\mathbf{x}$.

3.3 Projection onto an affine subspace and decomposition of observations

Projection onto $\mathcal{L}_q(\bar{\mathbf{x}})$ is done in two steps: (1) translate $\mathbf{x} \mapsto \mathbf{x} - \bar{\mathbf{x}}$ and project onto $\mathcal{L}_q$; (2) translate back by $\bar{\mathbf{x}}$:

$$P_{\mathcal{L}_q(\bar{\mathbf{x}})}(\mathbf{x}) = \bar{\mathbf{x}} + \mathbf{P_V}(\mathbf{x} - \bar{\mathbf{x}}) = \bar{\mathbf{x}} + s_{\mathbf{v}_1}(\mathbf{x} - \bar{\mathbf{x}})\mathbf{v}_1 + \cdots + s_{\mathbf{v}_q}(\mathbf{x} - \bar{\mathbf{x}})\mathbf{v}_q.$$

Hence an observation splits into projection plus remainder, and by Pythagoras

$$\mathbf{x} = \bar{\mathbf{x}} + \sum_{j=1}^{q} s_{\mathbf{v}_j}(\mathbf{x} - \bar{\mathbf{x}})\mathbf{v}_j + r_{\mathcal{L}_q}(\mathbf{x}), \qquad d(\mathbf{x}, \bar{\mathbf{x}})^2 = \sum_{j=1}^{q} s_{\mathbf{v}_j}(\mathbf{x} - \bar{\mathbf{x}})^2 + \|r_{\mathcal{L}_q}(\mathbf{x})\|^2.$$

This decomposition is the starting point of dimension reduction (PCA).

Remark 1.8: geometric meaning of variance

Projection scores are tied to conditional variances. If $X_1 \sim N(0, \sigma_1^2)$ and $X_2 = X_1 + \epsilon_2$ with independent $\epsilon_2 \sim N(0, \sigma_2^2)$, then $\text{Var}(X_2 \mid X_1) = \sigma_2^2$ and

$$\text{Var}(X_1 \mid X_2) = \sigma_1^2 - \frac{\sigma_1^4}{\sigma_1^2 + \sigma_2^2} = \frac{\sigma_1^2\sigma_2^2}{\sigma_1^2 + \sigma_2^2}$$

(conditional distribution formula, Ch. 3): what remains after removing the projected part is the conditional variance.

3.4 Linear and affine transformations

A $q \times p$ matrix $\mathbf{L}$ defines the linear transformation $\mathbf{x} \mapsto \mathbf{L}\mathbf{x}$; $q > p$ only adds dimensions without information and is rarely used. Examples: the projection $\mathbf{x} \mapsto \mathbf{P}_{\mathbf{V}}\mathbf{x}$ and the score map $\mathbf{x} \mapsto \mathbf{S}_{\mathbf{V}}\mathbf{x}$, $\mathbf{S}_{\mathbf{V}} = (\mathbf{v}_1/\|\mathbf{v}_1\|, \ldots, \mathbf{v}_q/\|\mathbf{v}_q\|)^\top$. Projection onto an affine subspace is not linear (because of the translation) but an **affine transformation** $\mathbf{x} \mapsto \boldsymbol{m} + \mathbf{L}\mathbf{x}$.

Theorem 1.9: affine maps and standardisation

1. Orthogonal projection onto $\mathcal{L}_q(\bar{\mathbf{x}})$ is affine with $\boldsymbol{m} = \bar{\mathbf{x}} - \mathbf{P}_{\mathbf{V}}\bar{\mathbf{x}}$, $\mathbf{L} = \mathbf{P}_{\mathbf{V}} = \mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top$; so is the score map $\mathbf{x} \mapsto (s_{\mathbf{v}_1}(\mathbf{x} - \bar{\mathbf{x}}), \ldots, s_{\mathbf{v}_q}(\mathbf{x} - \bar{\mathbf{x}}))^\top$.
2. (Standardisation) The data $\mathbf{x}'_i = \mathbf{D}^{-1/2}(\mathbf{x}_i - \bar{\mathbf{x}})$ have sample mean $\mathbf{0}$ and unit sample variances, and $\mathbf{x}' = \mathbf{D}^{-1/2}\mathbf{x} - \mathbf{D}^{-1/2}\bar{\mathbf{x}}$ is affine.

Proof. (1) $P_{\mathcal{L}_q(\bar{\mathbf{x}})}(\mathbf{x}) = \bar{\mathbf{x}} + \mathbf{P}_{\mathbf{V}}(\mathbf{x} - \bar{\mathbf{x}}) = (\bar{\mathbf{x}} - \mathbf{P}_{\mathbf{V}}\bar{\mathbf{x}}) + \mathbf{P}_{\mathbf{V}}\mathbf{x}$; the score vector is $\mathbf{S}_{\mathbf{V}}\mathbf{x} - \mathbf{S}_{\mathbf{V}}\bar{\mathbf{x}}$. (2) $\bar{\mathbf{x}}' = \frac{1}{n}\sum_i \mathbf{D}^{-1/2}(\mathbf{x}_i - \bar{\mathbf{x}}) = \mathbf{0}$; the jth sample variance is $s_{jj}^{-1/2} s_{jj} s_{jj}^{-1/2} = 1$. ■

3.5 Rotation of coordinates

For $p = 2$ and $\mathbf{R}_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$, put $\mathbf{v}_1 = \mathbf{R}_\theta\mathbf{e}_1$, $\mathbf{v}_2 = \mathbf{R}_\theta\mathbf{e}_2$; then $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2]$ satisfies $\mathbf{V}^\top\mathbf{V} = \mathbf{V}\mathbf{V}^\top = \mathbf{I}_2$. Since $\mathbf{x} = \mathbf{V}\mathbf{V}^\top\mathbf{x} = [\mathbf{v}_1, \mathbf{v}_2](\mathbf{v}_1^\top\mathbf{x}, \mathbf{v}_2^\top\mathbf{x})^\top$, the coordinates of $\mathbf{x}$ in the new system are the projection scores $(\mathbf{v}_1^\top\mathbf{x}, \mathbf{v}_2^\top\mathbf{x})$. A linear map by an orthonormal basis is a rotation of coordinates.

4 Standardisation, Sphering and Mahalanobis Distance

4.1 Univariate standardisation and its limitation

For $N(\mu, \sigma^2)$ the distance of x from the centre is better measured by the **standardised distance** $|x - \mu|/\sigma$ than by $|x - \mu|$. In the multivariate case, standardising each variable, $\mathbf{x} \mapsto \mathbf{D}^{-1/2}(\mathbf{x} - \bar{\mathbf{x}})$, removes scale differences but **ignores correlation**, so it does not give a distance that reflects the distribution.

Example 1.10: distances in bivariate data

$\bar{\mathbf{x}} = \mathbf{0}$, $\mathbf{S} = \begin{pmatrix} 640000 & 9600 \\ 9600 & 225 \end{pmatrix} = \begin{pmatrix} 800^2 & 800 \cdot 15 \cdot 0.8 \\ 800 \cdot 15 \cdot 0.8 & 15^2 \end{pmatrix}$ (correlation 0.8). For $\mathbf{y} = (-500, 10)^\top$, $\mathbf{z} = (1500, 15)^\top$: $d(\bar{\mathbf{x}}, \mathbf{y}) = 500.1$, $d(\bar{\mathbf{x}}, \mathbf{z}) = 1500.075$, so $\mathbf{z}$ looks farther. After standardisation $\mathbf{y}' = (-0.625, 0.667)^\top$, $\mathbf{z}' = (1.875, 1.000)^\top$ and still $d(\mathbf{0}, \mathbf{y}') = 0.91 < 2.125 = d(\mathbf{0}, \mathbf{z}')$. Yet the scatterplot has many points near $\mathbf{z}$ and almost none near $\mathbf{y}$ ($\mathbf{y}$ goes against the positive correlation). After standardisation the SD in the direction of $\mathbf{y}'$ is 0.36 and in the direction of $\mathbf{z}'$ is 1.31, not 1: standardisation does not standardise spread in every direction.

4.2 Sphering

If the covariance is $\mathbf{I}_p$, the data ellipse is a sphere ($c = \|\mathbf{x}-\bar{\mathbf{x}}\|^2$) and spread is 1 in **every** direction.

Theorem 1.11: isotropy of sphered data

If $\bar{\mathbf{x}} = \mathbf{0}$ and $\mathbf{S} = \mathbf{I}_p$, then for any unit vector $\mathbf{v}$ the scores $S_{\mathbf{v}}(\mathbf{x}_i) = \mathbf{v}^\top \mathbf{x}_i$ have sample variance $\mathbf{v}^\top \mathbf{S}\mathbf{v} = \mathbf{v}^\top \mathbf{v} = 1$.

Definition 1.12: sphering transformation

A transformation after which the covariance is $\mathbf{I}_p$. For general $\mathbf{S}$ it is built in three steps.

1. **Rotation**: choose orthogonal $\mathbf{U}_S = [\mathbf{u}_1, \dots, \mathbf{u}_p]$ so that $\mathbf{x}_i' = \mathbf{U}_S^\top \mathbf{x}_i$ has diagonal sample covariance $\mathbf{D}_S = \mathbf{U}_S^\top \mathbf{S}\mathbf{U}_S = \operatorname{diag}(d_{11}, \dots, d_{pp})$ (the eigendecomposition of $\mathbf{S}$).
2. **Standardisation**: $\mathbf{x}' \mapsto \mathbf{D}_S^{-1/2}\mathbf{x}'$.
3. **Back-rotation**: $\mathbf{x} \mapsto \mathbf{U}_S \mathbf{x}$.

Altogether $\mathbf{x} \mapsto \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top \mathbf{x} = \mathbf{S}^{-1/2}\mathbf{x}$.

The sphered covariance is $\mathbf{I}_p$.

$$\begin{aligned}\mathbf{S}' &= \tfrac{1}{n-1}\textstyle\sum_i \mathbf{S}^{-1/2}(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{S}^{-1/2} = \mathbf{S}^{-1/2}\mathbf{S}\mathbf{S}^{-1/2} \\ &= \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top \, \mathbf{U}_S \mathbf{D}_S \mathbf{U}_S^\top \, \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top = \mathbf{U}_S \mathbf{U}_S^\top = \mathbf{I}_p.\end{aligned}$$

■

Example 1.13: sphering in practice

(a) $\bar{\mathbf{x}} = (10, 10)^\top$, $\mathbf{S} = \operatorname{diag}(4, 9)$: the sphering matrix is $\mathbf{D}^{-1/2} = \operatorname{diag}(1/2, 1/3)$ and $\mathbf{S}' = \mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2} = \mathbf{I}_2$.

(b) $\mathbf{S} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$. With $\mathbf{U}_S = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ (rotation by $\theta = \pi/4$),

$$\mathbf{U}_S^\top \mathbf{S}\mathbf{U}_S = \tfrac{1}{2}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 9 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{D}_S,$$

so sphering is $\mathbf{x} \mapsto \mathbf{U}_S \operatorname{diag}(1/3, 1)\mathbf{U}_S^\top \mathbf{x}$. Applied to the data of Example 1.10, $\|\mathbf{y}' - \bar{\mathbf{x}}'\| = 2.65 > 2.18 = \|\mathbf{z}' - \bar{\mathbf{x}}'\|$: now $\mathbf{y}$ is correctly farther.

4.3 Mahalanobis distance

Definition 1.14: Mahalanobis (statistical) distance

For a distribution with mean $\bar{\mathbf{x}}$ and covariance $\mathbf{S}$, the Mahalanobis distance of $\mathbf{y}$ from $\bar{\mathbf{x}}$ is the Euclidean distance after sphering:

$$d_S(\mathbf{y}, \bar{\mathbf{x}}) = \|\mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top (\mathbf{y} - \bar{\mathbf{x}})\|, \qquad d_S(\mathbf{y}, \bar{\mathbf{x}})^2 = (\mathbf{y} - \bar{\mathbf{x}})^\top \mathbf{S}^{-1} (\mathbf{y} - \bar{\mathbf{x}}).$$

Proof. $d_S^2 = (\mathbf{y} - \bar{\mathbf{x}})^\top \mathbf{U}_S \mathbf{D}_S^{-1/2} \underbrace{\mathbf{U}_S^\top \mathbf{U}_S}_{\mathbf{I}} \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top (\mathbf{y} - \bar{\mathbf{x}})$ and $\mathbf{U}_S \mathbf{D}_S^{-1} \mathbf{U}_S^\top = \mathbf{S}^{-1}$. ■

Sphering and Mahalanobis distance are the multivariate counterparts of studentisation and standardised distance.

5 Eigendecomposition and Positive Definiteness

5.1 Eigendecomposition

For symmetric $\mathbf{A}$, $\mathbf{Au} = \lambda\mathbf{u}$ ($\mathbf{u} \neq \mathbf{0}$) makes $\mathbf{u}$ an eigenvector and λ an eigenvalue: $\mathbf{A}$ only stretches or shrinks $\mathbf{u}$, by the factor λ.

Theorem 1.15: eigendecomposition of a symmetric matrix

A symmetric $\mathbf{A} \in \mathbb{R}^{p\times p}$ satisfies $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top = \sum_{i=1}^{p} \lambda_i \mathbf{u}_i \mathbf{u}_i^\top$ with $\mathbf{U} = [\mathbf{u}_1, \ldots, \mathbf{u}_p]$ orthogonal and $\mathbf{\Lambda} = \operatorname{diag}(\lambda_1, \ldots, \lambda_p)$ real.

- If $\mathbf{u}_i$ is an eigenvector so is $c\mathbf{u}_i$; normalised to length 1 it is unique up to sign (for distinct eigenvalues).
- Eigenvectors of distinct eigenvalues are orthogonal.
- With repeated eigenvalues the decomposition exists but $\mathbf{U}$ is not unique, e.g. $\sigma^2\mathbf{I}_p = \mathbf{U}(\sigma^2\mathbf{I})\mathbf{U}^\top$ for every orthogonal $\mathbf{U}$.

Orthogonality. From $\mathbf{Au}_j = \lambda_j\mathbf{u}_j$, $\mathbf{u}_i^\top\mathbf{Au}_j = \lambda_j\mathbf{u}_i^\top\mathbf{u}_j$; from $\mathbf{A}^\top = \mathbf{A}$, $\mathbf{u}_i^\top\mathbf{A} = \lambda_i\mathbf{u}_i^\top$, so $\mathbf{u}_i^\top\mathbf{Au}_j = \lambda_i\mathbf{u}_i^\top\mathbf{u}_j$. Subtracting, $(\lambda_i - \lambda_j)\mathbf{u}_i^\top\mathbf{u}_j = 0$. ■

Example 1.16: eigenstructure of the sphering matrix

$\mathbf{A} = \mathbf{S}^{-1/2} = \begin{pmatrix} 2/3 & -1/3 \\ -1/3 & 2/3 \end{pmatrix}$ (Example 1.13(b)), $\mathbf{u}_1 = (1,1)^\top/\sqrt{2}$: $\mathbf{Au}_1 = \frac{1}{\sqrt{2}}(2/3 - 1/3, -1/3 + 2/3)^\top = \frac{1}{3}\mathbf{u}_1$; similarly $\mathbf{Au}_2 = \mathbf{u}_2$. Sphering rescales each eigen-direction by $1/\sqrt{\lambda_j}$.

5.2 Inverse, square root, definiteness

If no eigenvalue is 0, $\mathbf{A}^{-1} = \mathbf{U}\mathbf{\Lambda}^{-1}\mathbf{U}^\top$.

Definition 1.17: NND and PD

A symmetric $\mathbf{A}$ is **non-negative definite (NND)** if $\mathbf{x}^\top\mathbf{Ax} \geq 0$ for all $\mathbf{x}$, and **positive definite (PD)** if $\mathbf{x}^\top\mathbf{Ax} > 0$ for all $\mathbf{x} \neq \mathbf{0}$.

Theorem 1.18: eigenvalues and definiteness

A symmetric $\mathbf{A}$ is PD iff all eigenvalues are positive, and NND iff all are non-negative.

Proof. With $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$ and $\mathbf{y} = \mathbf{U}^\top\mathbf{x}$ (so $\mathbf{y} = \mathbf{0} \iff \mathbf{x} = \mathbf{0}$), $\mathbf{x}^\top\mathbf{Ax} = \mathbf{y}^\top\mathbf{\Lambda}\mathbf{y} = \sum_i \lambda_i y_i^2$. If all $\lambda_i > 0$ and $\mathbf{x} \neq \mathbf{0}$ some $y_i \neq 0$ and the sum is positive; NND is the same. Conversely, if $\mathbf{A}$ is PD, $\lambda_i = \mathbf{u}_i^\top\mathbf{Au}_i > 0$. ■

Theorem 1.19: the sample covariance is always NND, not necessarily PD

Proof. $\mathbf{z}^\top \mathbf{S}\mathbf{z} = \frac{1}{n-1}\sum_i \{(\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{z}\}^2 \geq 0$. But if the data lie in a lower-dimensional subspace of $\mathbb{R}^p$ (e.g. on a line), any $\mathbf{z} \neq \mathbf{0}$ orthogonal to it gives $\mathbf{z}^\top \mathbf{S}\mathbf{z} = 0$. ■

When n is much larger than p, $\mathbf{S}$ is PD in most cases and $\mathbf{S}^{-1}$ exists.

Definition 1.20: symmetric square root

For NND $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$, $\mathbf{S}^{1/2} = \mathbf{U}\mathbf{\Lambda}^{1/2}\mathbf{U}^\top$; for PD $\mathbf{S}$, $\mathbf{S}^{-1/2} = (\mathbf{S}^{-1})^{1/2} = \mathbf{U}\mathbf{\Lambda}^{-1/2}\mathbf{U}^\top$. $\mathbf{S}^{1/2}$ is symmetric with $\mathbf{S}^{1/2}\mathbf{S}^{1/2} = \mathbf{S}$.

5.3 Covariance matrices and ellipses

With $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$ the ellipse $E_S = \{\mathbf{x} : \mathbf{x}^\top \mathbf{S}^{-1}\mathbf{x} = 1\}$ reads

$$\frac{(\mathbf{u}_1^\top \mathbf{x})^2}{\lambda_1} + \cdots + \frac{(\mathbf{u}_p^\top \mathbf{x})^2}{\lambda_p} = 1,$$

so its principal axes are along $\mathbf{u}_i$ with half-lengths $\sqrt{\lambda_i}$. The eigendecomposition of a covariance matrix gives (1) the rotation $\mathbf{x} \mapsto \mathbf{U}^\top \mathbf{x}$ to uncorrelated coordinates and (2) the sphering $\mathbf{x} \mapsto \mathbf{U}\mathbf{\Lambda}^{-1/2}\mathbf{U}^\top \mathbf{x}$.

Theorem 1.21: covariance of rotated data

If $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$, the rotated data $\mathbf{x}_i' = \mathbf{U}^\top \mathbf{x}_i$ have sample covariance $\mathbf{\Lambda}$.

Proof. $\frac{1}{n-1}\sum_i \mathbf{U}^\top(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{U} = \mathbf{U}^\top \mathbf{S}\mathbf{U} = \mathbf{\Lambda}$. ■

The eigenvalues of a covariance matrix are the variances of mutually uncorrelated rotated coordinates — this is PCA.

6 Geometry of Principal Component Analysis

With many variables, inspecting all pairwise scatterplots is inefficient. PCA finds, in order, linear combinations of the original variables that explain the most variation, and summarises the data by a few of them (details in Ch. 7).

6.1 Which direction to project on?

The scores $t_i(\mathbf{a}) = \mathbf{a}^\top(\mathbf{x}_i - \bar{\mathbf{x}})$ along a unit vector $\mathbf{a}$ have sample variance $\frac{1}{n-1}\sum_i t_i(\mathbf{a})^2 = \mathbf{a}^\top \mathbf{S}\mathbf{a}$.

Definition 1.22: principal component directions

First direction $\mathbf{u}_1 \in \arg\max_{\|\mathbf{a}\|=1} \mathbf{a}^\top \mathbf{S}\mathbf{a}$; later directions maximise variance subject to orthogonality to the earlier ones. The solutions are eigenvectors of $\mathbf{S}$.

$\mathbf{u}_1$ is the top eigenvector. With $\mathbf{z} = \mathbf{U}^\top \mathbf{a}$ ($\|\mathbf{z}\| = 1$), $\mathbf{a}^\top \mathbf{S}\mathbf{a} = \sum_j \lambda_j z_j^2 \leq \lambda_1$, with equality at $\mathbf{a} = \mathbf{u}_1$. Under $z_1 = 0$ the maximum is λ_2 at $\mathbf{u}_2$, and so on. ■

6.2 Principal component scores

With $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$, $\lambda_1 \geq \cdots \geq \lambda_p \geq 0$, the score matrix is $\mathbf{T} = \mathbf{X}_C\mathbf{U} = (t_{ij})$, $t_{ij} = (\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{u}_j$. Each column of $\mathbf{T}$ has mean 0 and $\frac{1}{n-1}\mathbf{T}^\top \mathbf{T} = \mathbf{U}^\top \mathbf{S}\mathbf{U} = \mathbf{\Lambda}$: the jth PC has sample variance λ_j and different PC scores are uncorrelated.

Example 1.23: scores in two subjects

Student	Math x_1	English x_2	PC1 score	PC2 score
A	55	65	-14.14	-7.07
B	65	55	-14.14	7.07
C	70	70	0	0
D	75	85	14.14	-7.07
E	85	75	14.14	7.07

$\bar{\mathbf{x}} = (70, 70)^\top$, $\mathbf{S} = \begin{pmatrix} 125 & 75 \\ 75 & 125 \end{pmatrix}$, $\lambda_1 = 200$, $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1,1)^\top$, $\lambda_2 = 50$, $\mathbf{u}_2 = \frac{1}{\sqrt{2}}(1,-1)^\top$. $t_{i1} = \{(x_{i1} - 70) + (x_{i2} - 70)\}/\sqrt{2}$ is the overall level; $t_{i2} = \{(x_{i1} - 70) - (x_{i2} - 70)\}/\sqrt{2}$ is the relative difference between the subjects. A and B share the same PC1 score, so PC1 alone cannot distinguish their relative strengths.

6.3 Dimension reduction and reconstruction error

For student E, $\mathbf{x}_E = (85, 75)^\top$ and $t_{E1} = 10\sqrt{2}$. Reconstructing from PC1 gives $\hat{\mathbf{x}}_E = (70, 70)^\top + 10\sqrt{2} \cdot \frac{1}{\sqrt{2}}(1,1)^\top = (80, 80)^\top$: the average level is kept, the difference between subjects is lost.

Theorem 1.24: reconstruction error

For the reconstruction $\hat{\mathbf{x}}_i = \bar{\mathbf{x}} + \sum_{j \le q} t_{ij} \mathbf{u}_j$ from the first q PCs,

$$\sum_{i=1}^{n} \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2 = (n-1) \sum_{j=q+1}^{p} \lambda_j.$$

Among all q-dimensional planes through $\bar{\mathbf{x}}$, PCA minimises the reconstruction error, which is equivalent to maximising the total projected variance.

Proof. $\mathbf{x}_i - \hat{\mathbf{x}}_i = \sum_{j>q} t_{ij} \mathbf{u}_j$, so $\|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2 = \sum_{j>q} t_{ij}^2$ and $\sum_i t_{ij}^2 = (n-1)\lambda_j$. For any orthonormal $\mathbf{V} \in \mathbb{R}^{p \times q}$, Pythagoras gives $\sum_i \|(\mathbf{x}_i - \bar{\mathbf{x}}) - \mathbf{V}\mathbf{V}^\top(\mathbf{x}_i - \bar{\mathbf{x}})\|^2 = (n-1)\{\mathrm{tr}\mathbf{S} - \mathrm{tr}(\mathbf{V}^\top \mathbf{S} \mathbf{V})\}$, so minimising the error is maximising $\mathrm{tr}(\mathbf{V}^\top \mathbf{S} \mathbf{V})$, whose maximum $\lambda_1 + \cdots + \lambda_q$ follows from von Neumann's inequality (Ch. 2) or Ch. 7. ■

6.4 Proportion of variance, scale and caveats

- Since $\mathrm{tr}\mathbf{S} = \sum_k \lambda_k$, the proportion of variance explained by PC j is $\lambda_j / \sum_k \lambda_k$ and the cumulative proportion is $\sum_{j \le q} \lambda_j / \sum_k \lambda_k$ (80% = 200/250 in the example). Choose q from the cumulative proportion or a scree plot.
- Covariance-based PCA depends on units (rescaling English to 1000 points multiplies its variance by 100). PCA of standardised data $z_{ik} = (x_{ik} - \bar{x}_k)/s_k$ is **correlation PCA**. If differences in variance are meaningful, standardising is inappropriate.
- High explained variance does not mean good group separation. With $Y \in \{-1, 1\}$ equally likely, $X_1 = 10Z$, $X_2 = Y$, $Z \sim N(0,1) \perp\!\!\!\perp Y$: $\mathbf{\Sigma} = \mathrm{diag}(100, 1)$ and PC1 explains $100/101 \approx 99\%$, but $X_1 \perp\!\!\!\perp Y$ while X_2 alone separates the groups perfectly. PCA keeps variation, not predictive information.
- Still, for the Golub data ($p = 7129$), the PC1–PC2 plot, built without the labels, separates ALL/AML better than the original variables: PCA can be used to *explore* group differences.

7 Random Vectors and the Geometry of Variable Vectors

7.1 Random vectors

The data $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_n]^\top$ are a random sample from a population modelled by the joint distribution of $\mathbf{x} = (X_1, \ldots, X_p)^\top \sim f$.

Example 1.25: distributions of random vectors

1. Independent $X_i \sim \text{Unif}(a_i, b_i)$: $f(\mathbf{x}) = \prod_i \frac{1}{b_i - a_i} \mathbf{1}\{x_i \in (a_i, b_i)\}$, uniform on a hypercube.
2. Independent $X_i \sim N(0,1)$: $f(\mathbf{x}) = (2\pi)^{-p/2} e^{-\|\mathbf{x}\|^2/2}$, the standard multivariate normal.
3. (X_1, X_2) uniform on $0 < x_1 < x_2 < 1$: $f = 2 \cdot \mathbf{1}\{0 < x_1 < x_2 < 1\}$. $X_1 \mid X_2 = x_2 \sim \text{Unif}(0, x_2)$ depends on x_2, so they are not independent.
4. Bivariate normal with correlation ρ: $f = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\{-\frac{x_1^2 - 2\rho x_1 x_2 + x_2^2}{2(1-\rho^2)}\}$. Both marginals are $N(0,1)$ but the joint density is not their product: not independent.

7.2 Mean, variance and correlation through variable vectors

Viewing the variable vectors $\mathbf{x}_{(j)} \in \mathbb{R}^n$ (p vectors in n-space) gives a geometric reading of the statistics.

- **Mean**: $\bar{x}_j = \frac{1}{n} \mathbf{1}_n^\top \mathbf{x}_{(j)}$. The projection of $\mathbf{x}_{(j)}$ onto $\mathbf{1}_n$ is $P_{\mathbf{1}_n}(\mathbf{x}_{(j)}) = \mathbf{1}_n (\mathbf{1}_n^\top \mathbf{1}_n)^{-1} \mathbf{1}_n^\top \mathbf{x}_{(j)} = \bar{x}_j \mathbf{1}_n$ with score $\sqrt{n}\,\bar{x}_j$.
- **Deviation vector**: $\mathbf{d}_j = \mathbf{x}_{(j)} - \bar{x}_j \mathbf{1}_n$; $\mathbf{x}_{(j)} = \bar{x}_j \mathbf{1}_n + \mathbf{d}_j$ is an orthogonal decomposition ($\mathbf{1}_n^\top \mathbf{d}_j = 0$).
- $s_j = \|\mathbf{d}_j\|/\sqrt{n-1}$, $s_j^2 = \|\mathbf{d}_j\|^2/(n-1)$, $s_{jk} = \mathbf{d}_j^\top \mathbf{d}_k/(n-1)$.
- **The sample correlation is the cosine of the angle between deviation vectors**: $r_{jk} = \dfrac{\mathbf{d}_j^\top \mathbf{d}_k}{\|\mathbf{d}_j\| \|\mathbf{d}_k\|} = \cos\theta_{jk}$.

A longer deviation vector means more variability. $\mathbf{d}_j = c\mathbf{d}_k$ with $c > 0$ ($\theta = 0$) gives $r = 1$, with $c < 0$ ($\theta = \pi$) gives $r = -1$; orthogonal deviation vectors give zero *linear* correlation.

Theorem 1.26: unbiasedness of $\bar{\mathbf{X}}$ and S

If $\mathbf{X}_1, \ldots, \mathbf{X}_n$ is a random sample with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$,

$$\mathrm{E}(\bar{\mathbf{X}}) = \boldsymbol{\mu}, \qquad \mathrm{Cov}(\bar{\mathbf{X}}) = \tfrac{1}{n}\boldsymbol{\Sigma}, \qquad \mathrm{E}(\mathbf{S}_n) = \tfrac{n-1}{n}\boldsymbol{\Sigma}, \qquad \mathrm{E}(\mathbf{S}) = \boldsymbol{\Sigma}.$$

Proof. $\mathrm{E}\bar{\mathbf{X}} = \frac{1}{n}\sum \mathrm{E}\mathbf{X}_j = \boldsymbol{\mu}$. $\mathrm{Cov}(\bar{\mathbf{X}}) = \frac{1}{n^2}\sum_j \sum_\ell \mathrm{E}\{(\mathbf{X}_j - \boldsymbol{\mu})(\mathbf{X}_\ell - \boldsymbol{\mu})^\top\}$, and terms with $j \neq \ell$ vanish by independence, giving $\frac{1}{n^2} \cdot n\boldsymbol{\Sigma}$. Since $\sum_j (\mathbf{X}_j - \bar{\mathbf{X}})(\mathbf{X}_j - \bar{\mathbf{X}})^\top = \sum_j (\mathbf{X}_j - \boldsymbol{\mu})(\mathbf{X}_j - \boldsymbol{\mu})^\top - n(\bar{\mathbf{X}} - \boldsymbol{\mu})(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top$, its expectation is $n\boldsymbol{\Sigma} - \boldsymbol{\Sigma} = (n-1)\boldsymbol{\Sigma}$. ■

7.3 Linear combinations of variables

For the new variable $z = a + \mathbf{b}^\top \mathbf{x} = a + \sum_j b_j x_j$ the variable vector is $\mathbf{z} = a\mathbf{1}_n + \sum_j b_j \mathbf{x}_{(j)}$, with

$$P_{\mathbf{1}_n}(\mathbf{z}) = \Big(a + \sum_j b_j \bar{x}_j\Big)\mathbf{1}_n, \qquad \mathbf{d}_z = \sum_j b_j \mathbf{d}_j = \mathbf{X}_C \mathbf{b}.$$

Theorem 1.27: sample variance and covariance of linear combinations

$$\widehat{\mathrm{Var}}(z) = \tfrac{1}{n-1}\mathbf{d}_z^\top \mathbf{d}_z = \mathbf{b}^\top \mathbf{S}\mathbf{b}, \qquad y = \mathbf{a}^\top \mathbf{x},\ z = \mathbf{b}^\top \mathbf{x} \ \Rightarrow\ \widehat{\mathrm{Cov}}(y,z) = \tfrac{1}{n-1}\mathbf{d}_y^\top \mathbf{d}_z = \mathbf{a}^\top \mathbf{S}\mathbf{b}.$$

For two variables, $y = b\,x_1 + c\,x_2$ has $\widehat{\mathrm{Var}}(y) = \{b^2\|\mathbf{d}_1\|^2 + c^2\|\mathbf{d}_2\|^2 + 2bc\|\mathbf{d}_1\|\|\mathbf{d}_2\|\cos\theta\}/(n-1)$.

8 Dimension and Rank of the Data

- Linear independence: $\sum c_i \mathbf{x}_i = \mathbf{0} \Rightarrow$ all $c_i = 0$. The rank is the maximal number of linearly independent columns (= rows). $\mathrm{rank}(\mathbf{X}) \le \min(n, p)$.
- $\mathrm{rank}(\mathbf{A}) = \mathrm{rank}(\mathbf{A}^\top) = \mathrm{rank}(\mathbf{A}^\top\mathbf{A})$, hence $\mathrm{rank}(\mathbf{S}) = \mathrm{rank}(\mathbf{X}_C^\top\mathbf{X}_C) = \mathrm{rank}(\mathbf{X}_C)$.
- All deviation vectors are orthogonal to $\mathbf{1}_n$, so they lie in an $(n-1)$-dimensional subspace: $\mathrm{rank}(\mathbf{X}_C) \le \min(n-1, p)$.
- If $p > n$, $\mathrm{rank}(\mathbf{S}) \le n - 1 < p$: $\mathbf{S}$ is always rank-deficient.

Theorem 1.28: rank-deficiency property

For the covariance matrix $\mathbf{S}$ of $\mathbf{X} \in \mathbb{R}^{n\times p}$ the following are equivalent.

1. $\mathbf{S}$ is rank-deficient.
2. There is $\mathbf{a} \neq \mathbf{0}$ with $\mathbf{a}^\top(\mathbf{x}_i - \bar{\mathbf{x}}) = 0$ for all i.
3. All projection scores coincide: $s_{\mathbf{a}}(\mathbf{x}_i) = s_{\mathbf{a}}(\mathbf{x}_j)$ for all i, j.
4. $\mathrm{span}\{\mathbf{d}_1, \ldots, \mathbf{d}_p\}$ has dimension less than p.
5. Some deviation vector is a linear combination of the others.
6. At least one eigenvalue of $\mathbf{S}$ is 0.

Proof. (1)⇔(2): $\mathrm{rank}(\mathbf{S}) = \mathrm{rank}(\mathbf{X}_C^\top) = r < p$ iff the columns $\{\mathbf{x}_i - \bar{\mathbf{x}}\}$ of $\mathbf{X}_C^\top$ span a space of dimension $r < p$, iff a nonzero $\mathbf{a}$ in its orthogonal complement exists. (2)⇔(3): $\mathbf{a}^\top(\mathbf{x}_i - \bar{\mathbf{x}}) = 0 \iff \mathbf{a}^\top\mathbf{x}_i = \mathbf{a}^\top\bar{\mathbf{x}}$ for all i. (1)⇔(4): $\mathrm{rank}(\mathbf{S}) = \mathrm{rank}(\mathbf{X}_C) = \dim \mathrm{span}\{\mathbf{d}_j\}$. (4)⇔(5): dimension below p means $\{\mathbf{d}_j\}$ is linearly dependent. (2)⇒(6): $\mathbf{S}\mathbf{a} = \frac{1}{n-1}\mathbf{X}_C^\top\mathbf{X}_C\mathbf{a} = \mathbf{0} = 0\cdot\mathbf{a}$. (6)⇒(2): $\mathbf{S}\mathbf{u} = \mathbf{0}$ gives $\mathbf{u}^\top\mathbf{S}\mathbf{u} = \frac{1}{n-1}\|\mathbf{X}_C\mathbf{u}\|^2 = 0$, so $\mathbf{X}_C\mathbf{u} = \mathbf{0}$. ■

If $\mathbf{S}$ is full rank, every direction carries independent variation. If it is rank-deficient, some direction carries no information at all (an exact linear relation among the variables), the sphering $\mathbf{x} \mapsto \mathbf{S}^{-1/2}\mathbf{x}$ is undefined, and statistical analysis becomes less reliable. In high dimensions ($p > n$) $\mathbf{S}$ is always rank-deficient: $\mathrm{rank}(\mathbf{S}) \le \min\{n-1, p\} < p$.